

Findable, accessible, interoperable, and reusable

Modeling the distribution of marine biodiversity under global climate change

Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal

Findable, Accessible, Interoperable, and Reusable

Making research more effective: Applying FAIR principles ensures that the data, code, and outputs associated with SDM studies are managed for maximum reuse and impact.

Findable: can others (and machines) find your data and models?

Accessible: once found, how can the data/model be accessed?

Interoperable: can the data be integrated and used with other data or workflows?

Reusable: can the data/model be effectively used in future studies?

Benefit: applying FAIR principles to inputs (data, code) and outputs (maps) enhances transparency, reproducibility, collaboration, and the overall value of the research.

Received: 30 July 2023 | Revised: 24 January 2024 | Accepted: 24 January 2024

DOI: 10.1111/geb.13813

DATA ARTICLE

Global Ecology
and Biogeography

A Journal of
Macroecology

WILEY

Bio-ORACLE v3.0. Pushing marine data layers to the CMIP6 Earth System Models of climate change research

Jorge Assis^{1,2}  | Salvador Jesús Fernández Bejarano³  | Vinícius W. Salazar⁴  |
Lennert Schepers³  | Lidiane Gouvêa¹  | Eliza Fragkopoulou¹  | Frederic Leclercq³  |
Bart Vanhoorne³  | Lennert Tyberghein³  | Ester A. Serrão¹  |
Heroen Verbruggen⁵  | Olivier De Clerck⁶

¹Centre of Marine Sciences (CCMAR/CIMAR LA), Universidade do Algarve, Faro, Portugal

²Faculty of Bioscience and Aquaculture, Nord Universitet, Bodø, Norway

³Flanders Marine Institute, Ostend, Belgium

⁴Melbourne Integrative Genomics, School of Mathematics and Statistics, University of Melbourne, Melbourne, Victoria, Australia

⁵School of BioSciences, University of Melbourne, Parkville, Victoria, Australia

⁶Department of Biology, Phycology Research Group, Ghent University, Ghent, Belgium

Correspondence

Jorge Assis, Centre of Marine Sciences (CCMAR/CIMAR LA), Universidade do Algarve, Faro, Portugal.

Abstract

Motivation: Impacts of climate change on marine biodiversity are often projected

Data in Brief 53 (2024) 110200



ELSEVIER

Contents lists available at [ScienceDirect](#)

journal homepage: [www.elsevier.com/locate/dib](#)

Data Article

A global dataset of demosp...

Data in Brief 48 (2023) 109223



ELSEVIER

Contents lists available at [ScienceDirect](#)

journal homepage: [www.elsevier.com/locate/dib](#)

Data Article

A dataset of cold-water coral distribution

[www.nature.com/scientificdata](#)

SCIENTIFIC DATA

OPEN

DATA DESCRIPTOR

A fine-tuned global distribution dataset of marine forests

Jorge Assis^{1,4}, Eliza Fragkopoulou^{1,4}, Duarte Frade¹, João Neiva¹, André Oliveira¹, David Abecasis¹, Sylvain Faugeron^{2,3} & Ester A. Serrão¹

Species distribution records are a prerequisite to follow climate-induced range shifts across space and time. However, synthesizing information from various sources such as peer-reviewed literature, herbaria, digital repositories and citizen science initiatives is not only costly and time consuming, but also challenging, as data may contain thematic and taxonomic errors and generally lack standardized

Eliza Fragkopoulou^{a,*}, Ester A. Serrão^a

MAR-CIMAR), University of the Algarve, 8005-139 Faro, Portugal
aculture, Nord Universitet, Postboks 1490, Bodø, Norway

Check for updates

ABSTRACT

Species distribution data are key for monitoring present and future biodiversity patterns and informing conservation and management strategies. Large biodiversity information facilities often contain spatial and taxonomic errors that reduce the quality of the provided data. Moreover, datasets are frequently shared in varying formats, inhibiting proper integration and interoperability. Here, we provide a quality-controlled dataset of the diversity and distribution of cold-water coral

Data in Brief 52 (2024) 110023



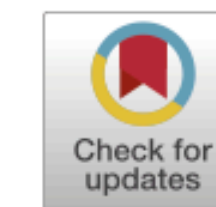
Contents lists available at ScienceDirect

Data in Brief

journal homepage: www.elsevier.com/locate/dib

Data Article

Range map data of marine ecosystem structuring species under global climate change



Lidiane Gouvêa^{a,*}, Eliza Fragkopoulou^a, T rence Legrand^a,
Ester A. Serr o^a, Jorge Assis^{a,b,*}

^a Centre of Marine Sciences, University of Algarve, Faro, Portugal

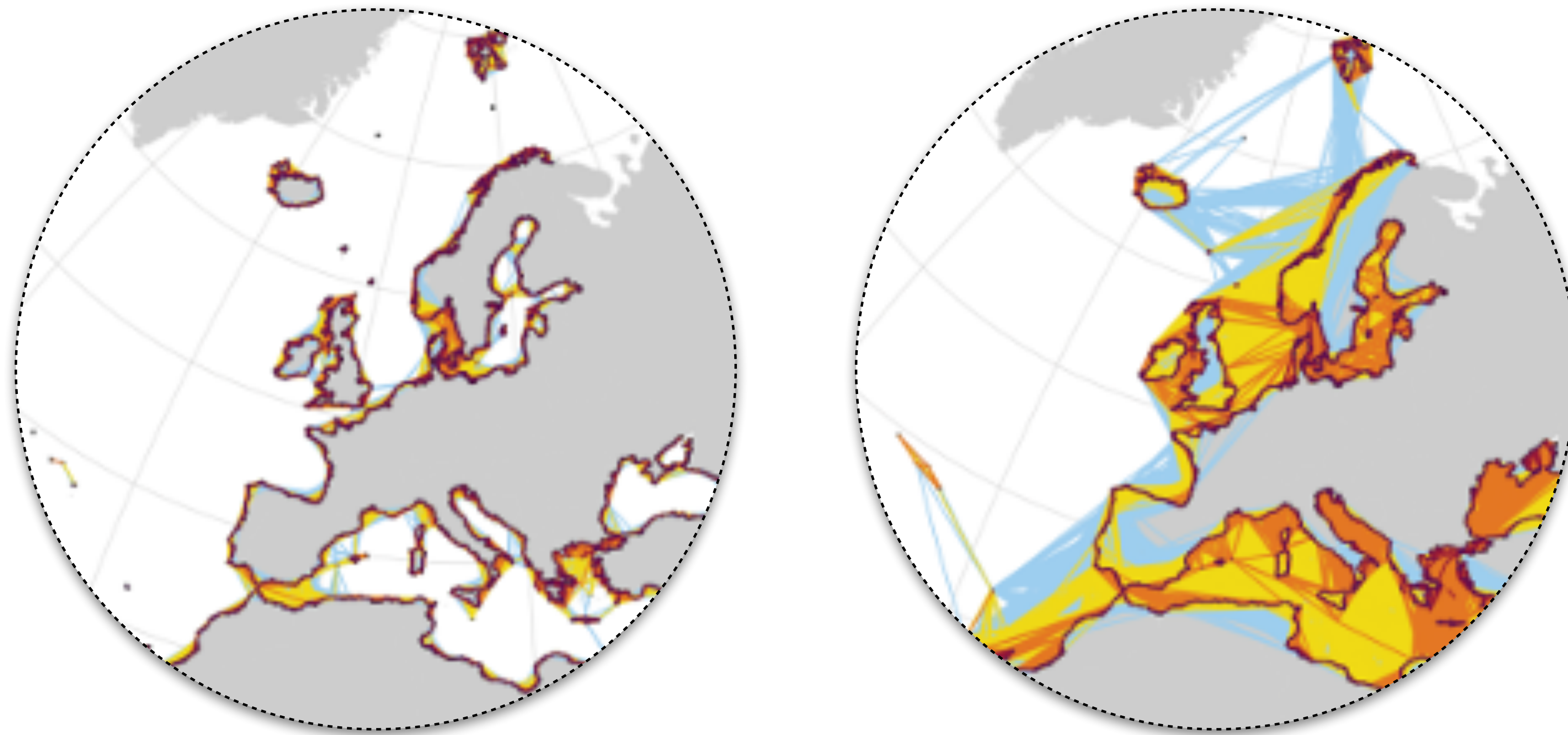
^b Faculty of Bioscience and Aquaculture, Nord Universitet, Bod , Norway

ARTICLE INFO

Article history:
Received 23 September 2023
Received in revised form 20 October 2023
Accepted 21 October 2023

ABSTRACT

Data on contemporary and future geographical distributions of marine species are crucial for guiding conservation and



github.com/jorgeassis/coastalNet

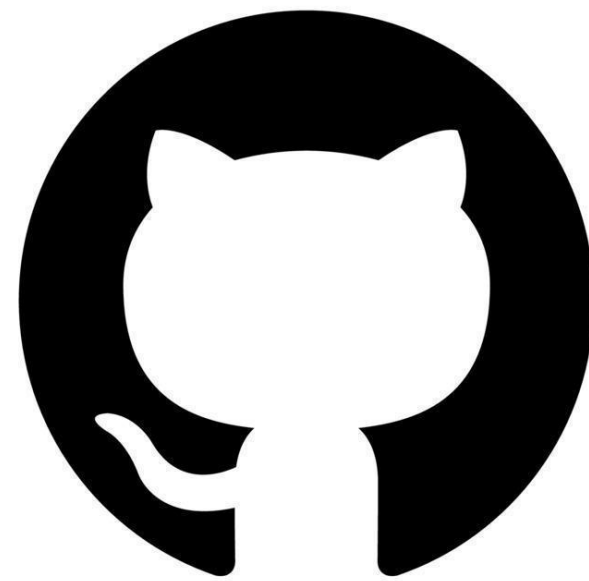
A dataset of oceanographic connectivity estimates along the world's coastlines derived from a biophysical model that assimilates 21-years of ocean current data.



Open access repositories

Online open access repositories allow researchers to **preserve and share their research outputs, including figures, datasets, images, and videos.**

E.g., Figshare and Zenodo are free to upload content and free to access, in adherence to the principle of open data.



GitHub

Git is an essential tool for managing complex analytical workflows like SDM, promoting reproducibility, collaboration, and saving us from losing work or getting lost in different versions of your scripts.

Git allows to access the exact version of the code used to generate specific results or maps, supporting transparency and validation (complements FAIR Principles).